Signals and Systems
SET 214
Lecture (11)- Sample Theory

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Outlines

- Sampling Theory
- Continuous time vs Discrete time
- Sampling Processing
- Sampling Theorem
- Sampling Theorem: Intuitive proof
- Discrete-Time Signal
- Exercise

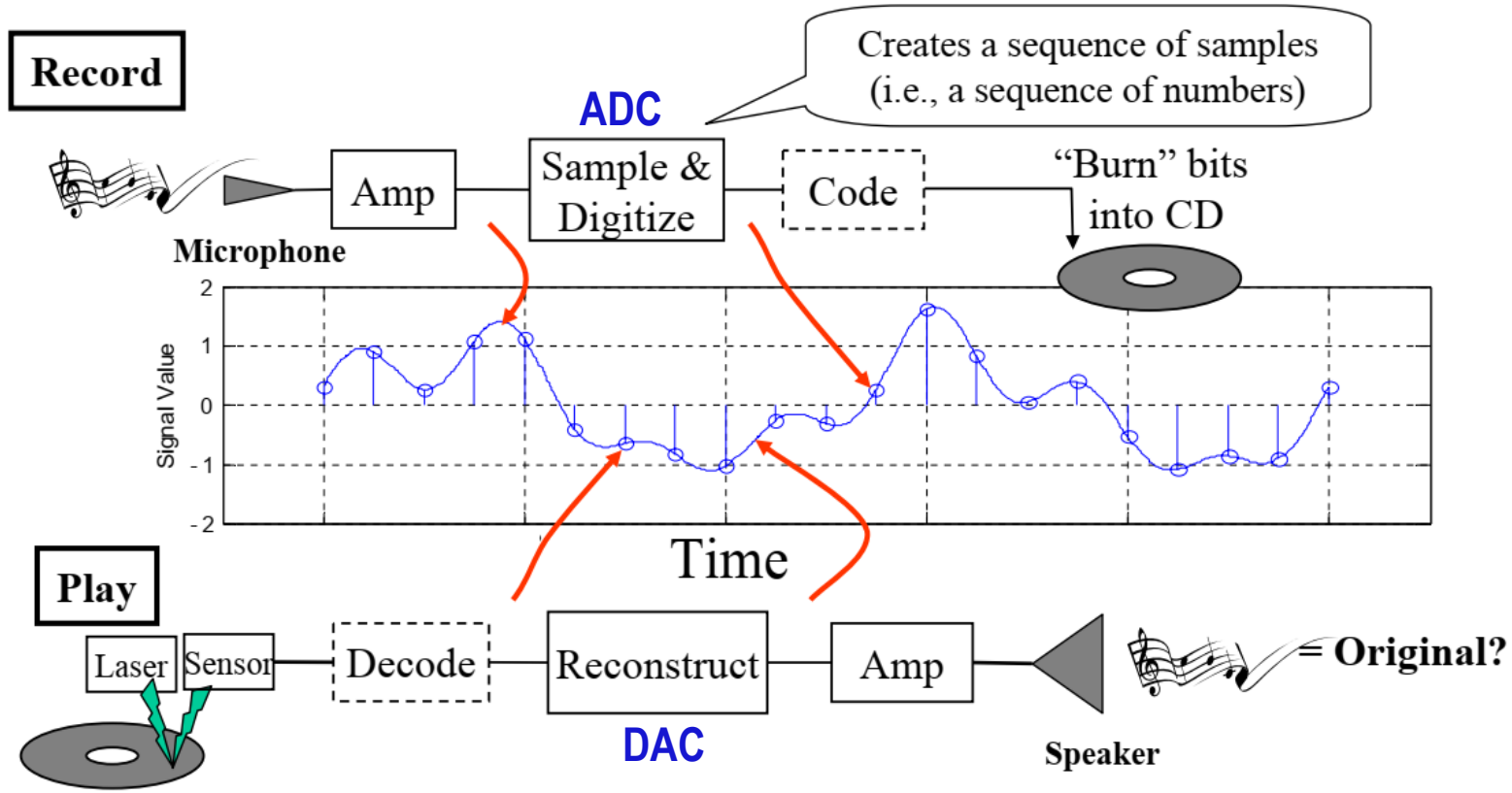
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Sampling Theory (1/2)

- **Sampling**: The process of converting a **continuous** time signal to **discrete** time signal so that computers can process the **data digitally**

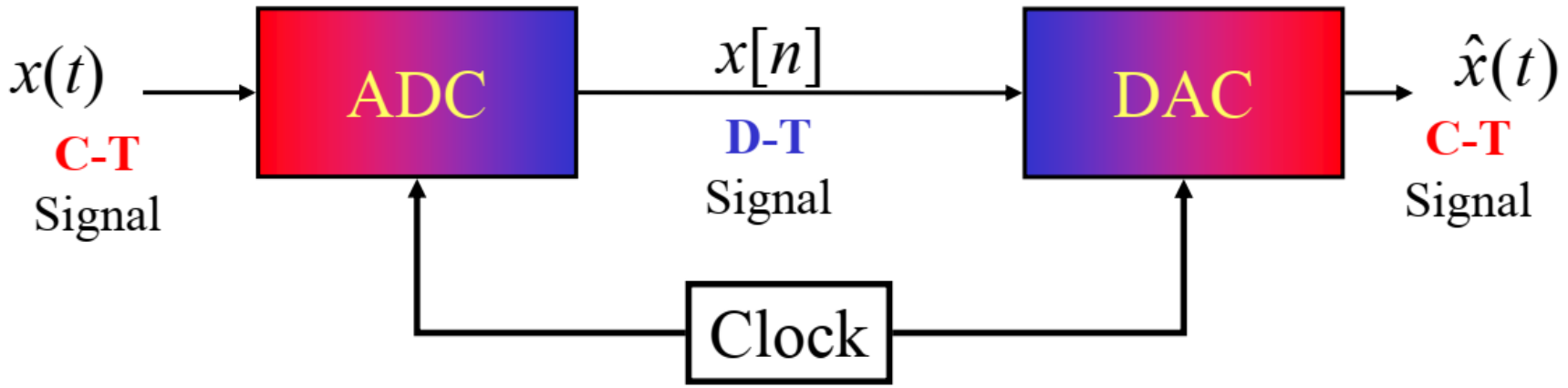
- Systems that use Digital Signal Proc. (DSP) generally
  - get a **continuous-time signal** from a sensor
  - a **continuous-time system** amplified the signal
  - an "analog-to-digital converter" (ADC) samples the signal to create a **discrete-time signal**.
  - A discrete-time system to do the Digital Signal Processing
  - and then (if desired) convert back to analog using a "digital to-analog converter (DAC)



Sampled & Digitized music on a Compact Disc

Sampling Theory (2/2)

- If Sampling is "Valid. We Should be able to "Perfectly" Reconstruct from Samples



Can we make: $\hat{x}(t) = x(t)$???

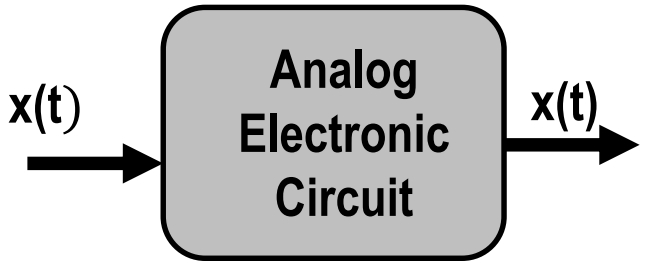
- **Aliasing** – the original signal is corrupted in a way that recovery is not possible.

Outlines

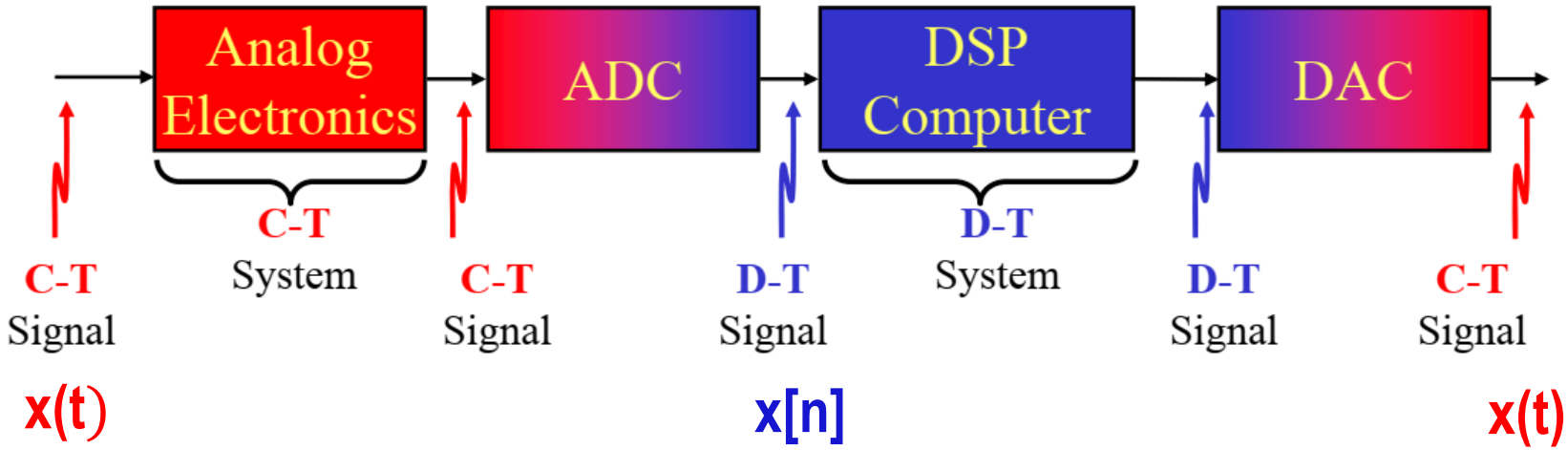
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Continuous time vs Discrete time

- **Continuous time system**
  - Good for analogue & general understanding
  - Appropriate mostly to analogue electronic systems



- **Discrete time system**
  - Electronics are increasingly digital. Ex., digital ASIC chips, FPGAs and microprocessors…etc



Sampling Process ???

Outlines

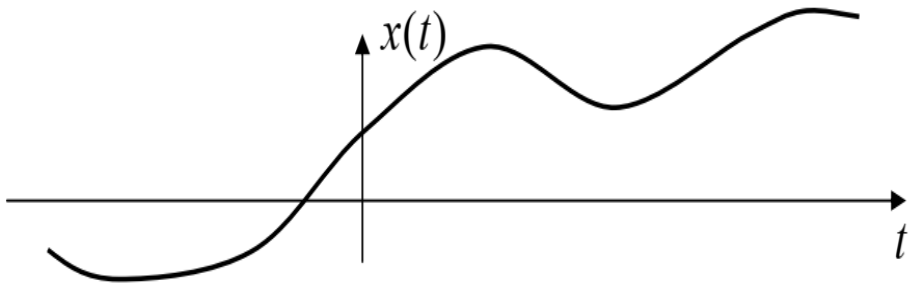
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Sampling Processing

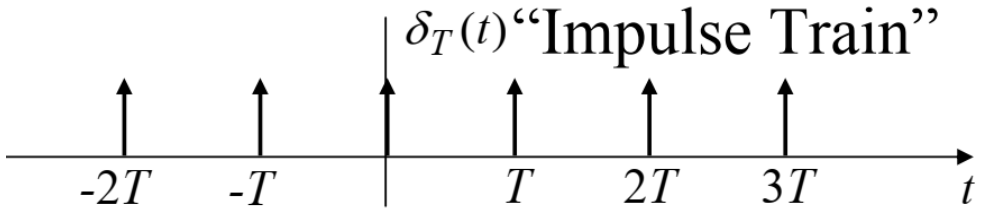
Take a sample every sampling period T_s – uniform sampling

$$T_s = \frac{1}{fs} \quad \Longleftrightarrow \quad f_s = \frac{1}{Ts}$$

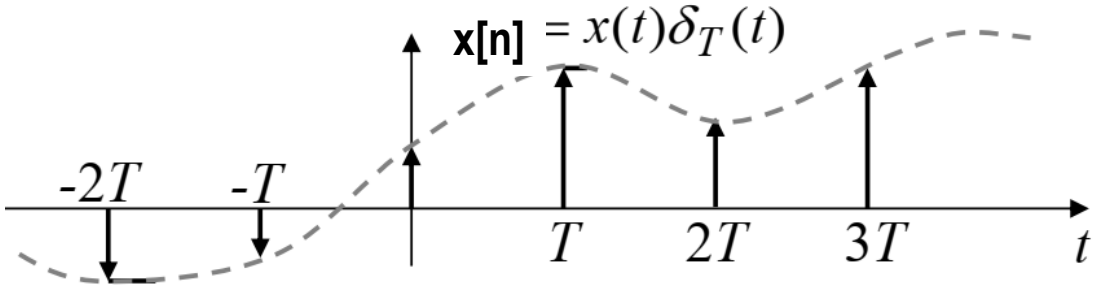




T=T_s







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Sampling Theorem

A continuous-time signal x(t) with frequencies no higher than f_{max} (Hz) can be reconstructed **EXACTLY** from its samples $x[n] = x(nT_s)$, if the samples are taken at a rate $f_s = \frac{1}{T_s}$ that is greater than $2*f_{max}$.

Sampling Theorem is sometimes known as the Shannon Theorem or the Nyquist Theorem

For example, the input signal x(t)= cos(2*pi*1000*t)→ signal frequency= 1000Hz

- **We need to sample this input signal at higher than 2000 Hz (i.e. 2000 samples per second) in order**
  - **NOT to lose any data,**
  - **Reconstruct the input signal exactly**

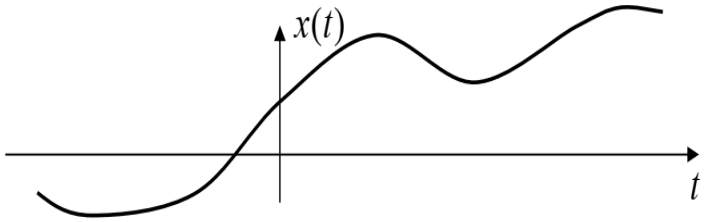
What if you ignore this rule? You will corrupt the signal by introducing spurious signals through a phenomenon known as "aliasing"

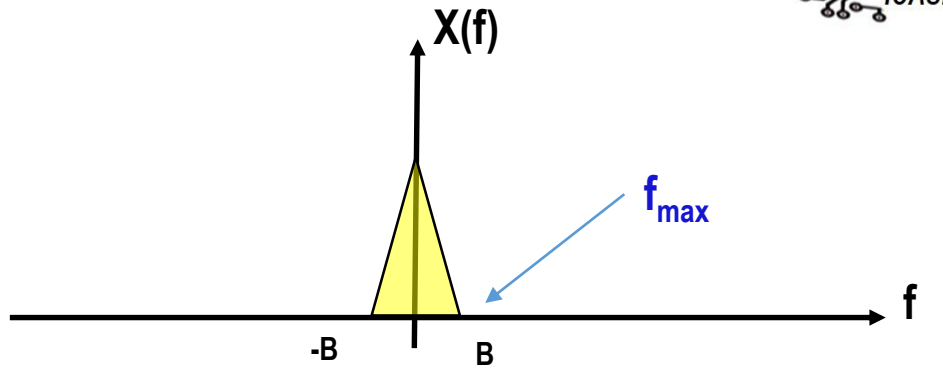
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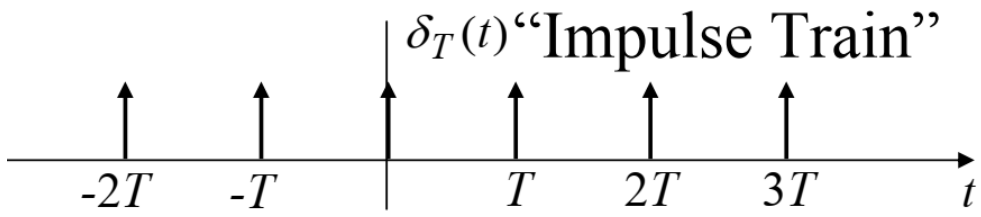
Sampling Theorem: Intuitive proof (1/5)

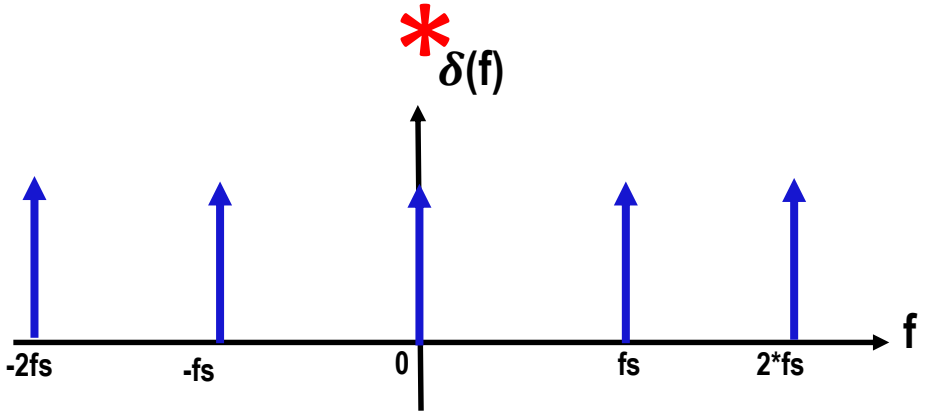
Consider a band limited signal x(t) is BW→B=f_{max}

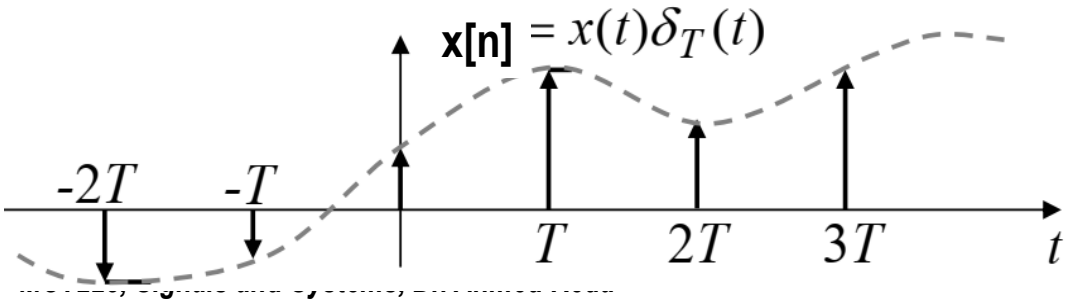




Note:
Multiplication in time domain is converted to convolution in frequency domain



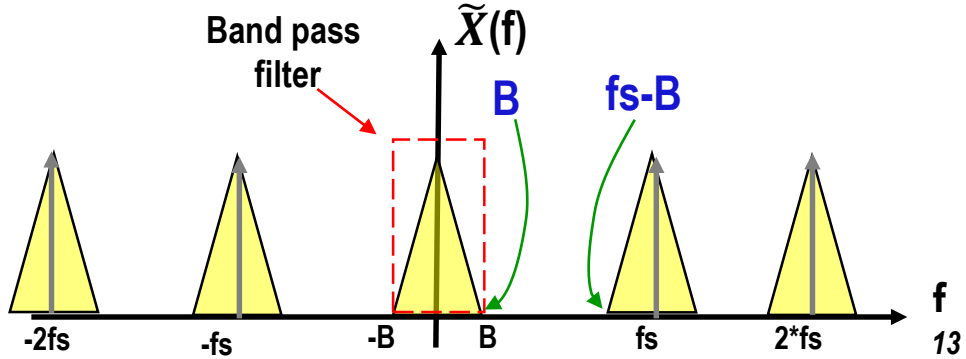




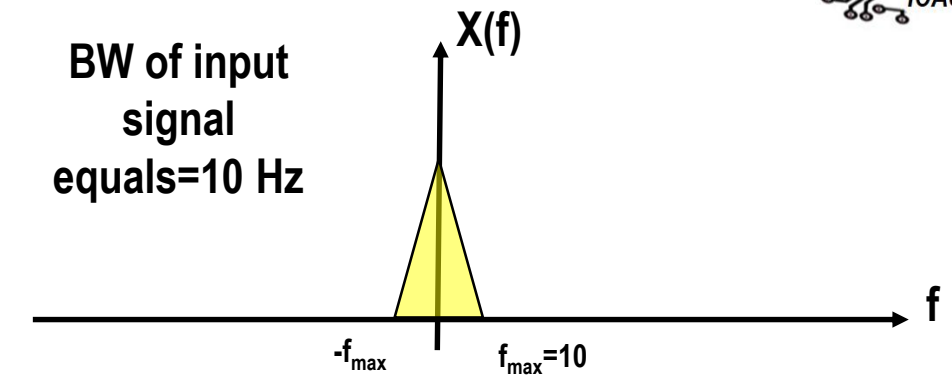
don't overlap the original

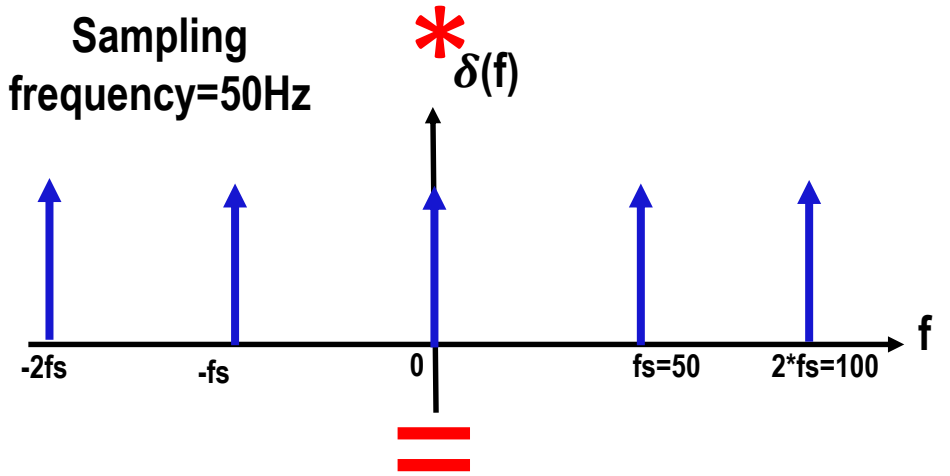
fs-B ≥B

fs ≥ 2B



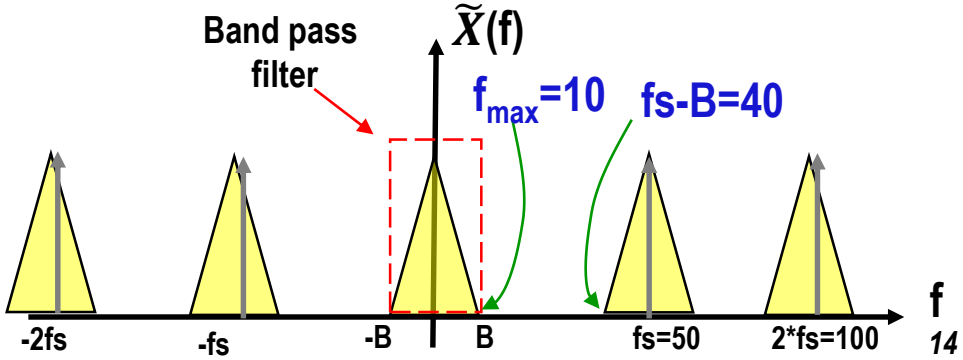
Sampling Theorem: Intuitive proof (2/5)





$f_s >> f_{max}$

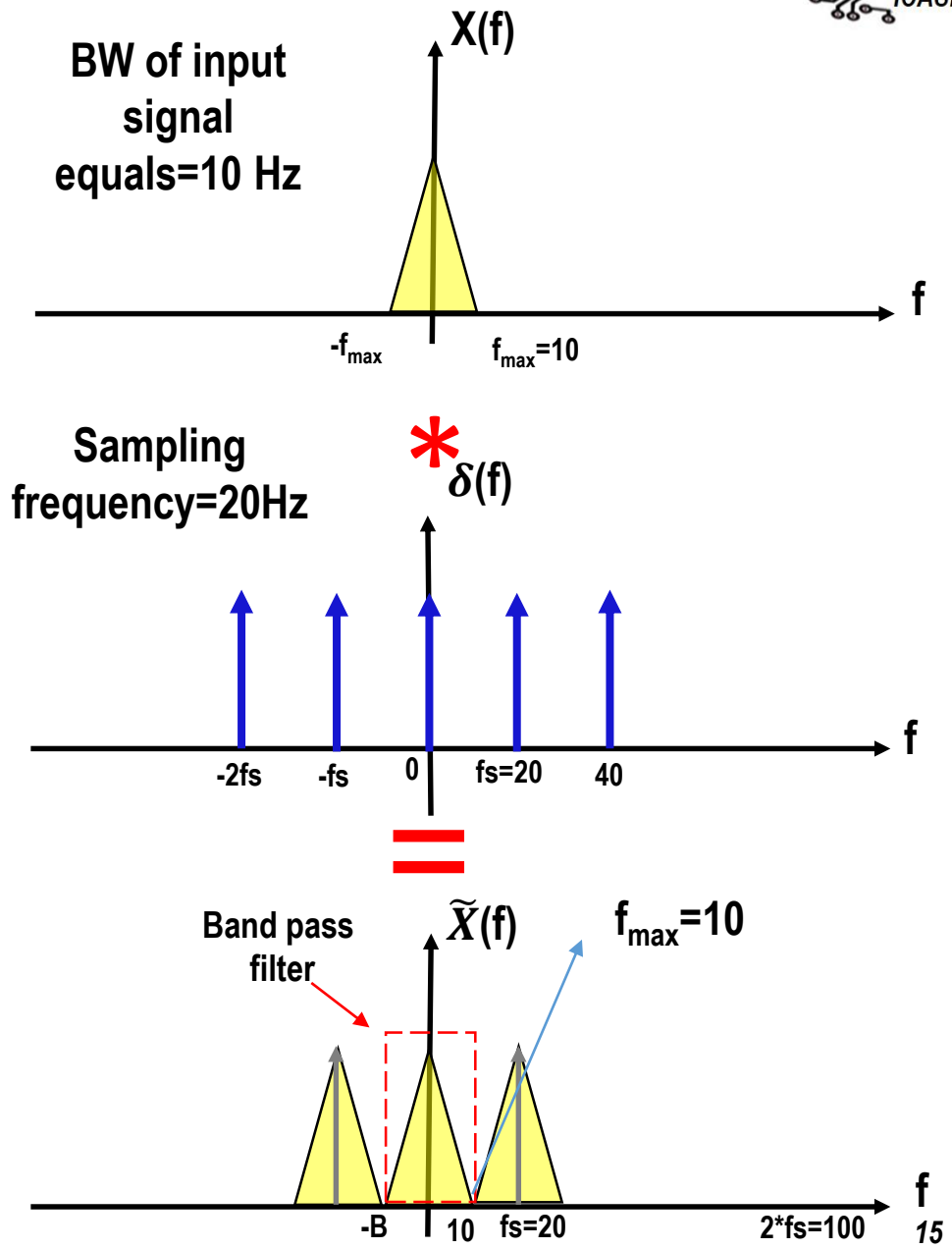
Perfect reconstruction possible



Sampling Theorem: Intuitive proof (3/5)

$$f_s = 2 * f_{max}$$

perfect reconstruction possible

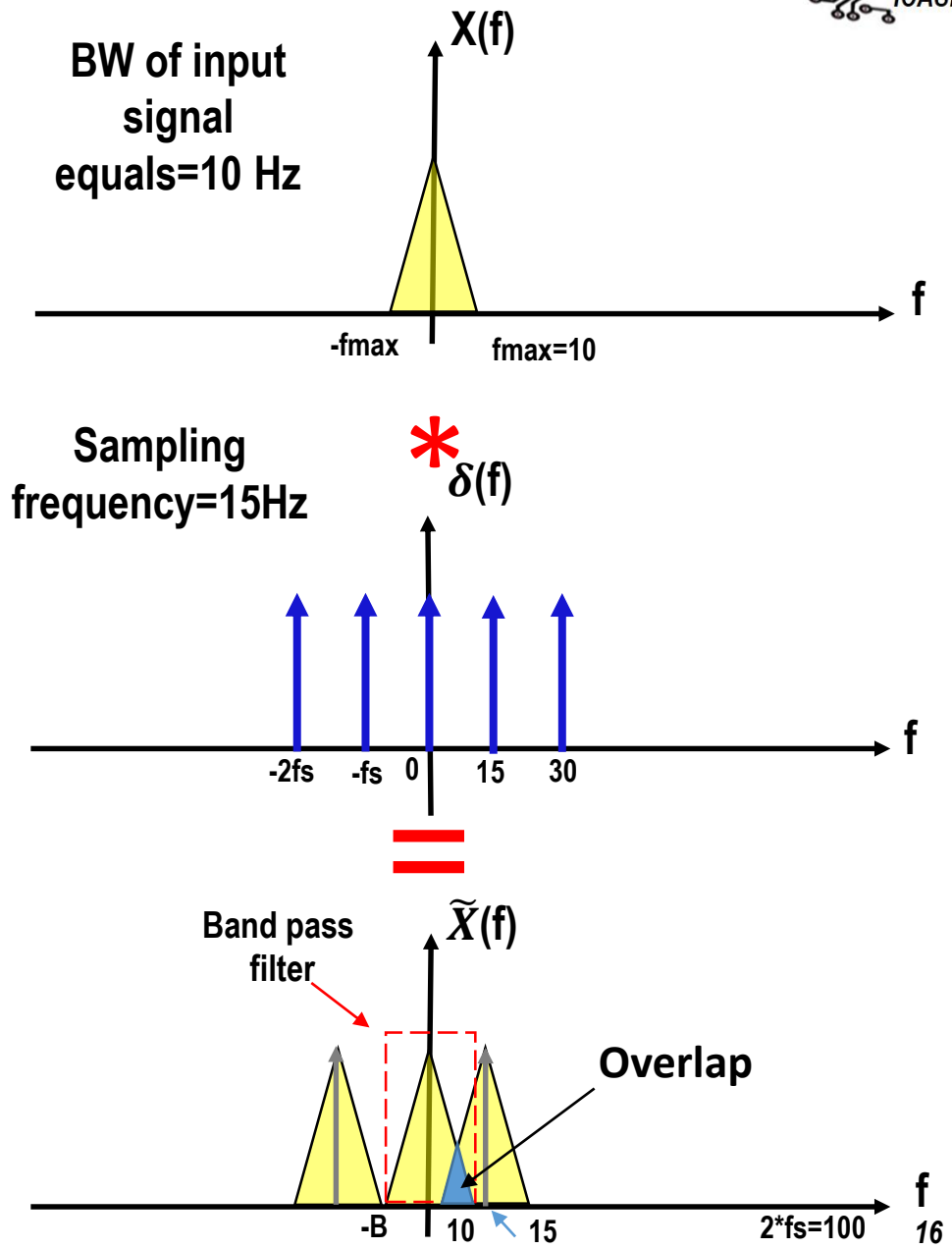


Sampling Theorem: Intuitive proof (4/5)

$f_s < 2 * f_{max}$

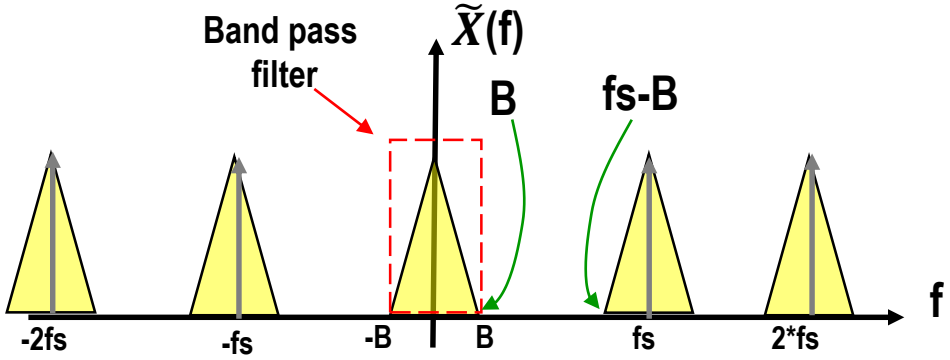
Aliasing

To avoid corruption of signal after sampling, one must ensure that $fs \geq 2 * f_{max}$



Sampling Theorem: Intuitive proof (5/5)

Therefore, to reconstruct the original signal x(t), we can use an ideal low pass filter on the sampled spectrum



don't overlap the original

$$fs-B \geq B$$

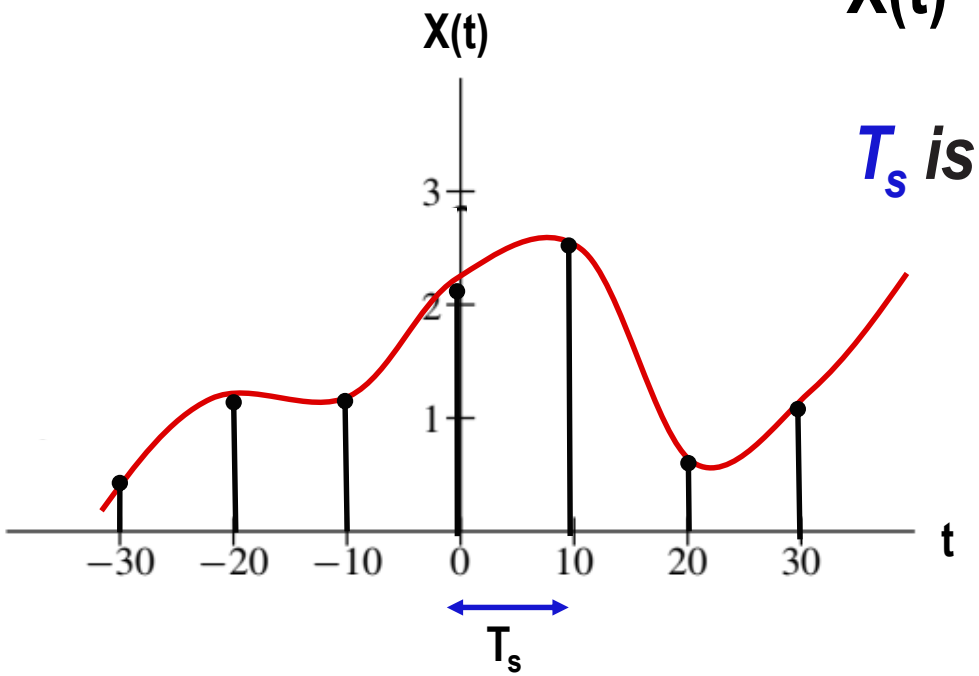
$$fs \geq 2B$$

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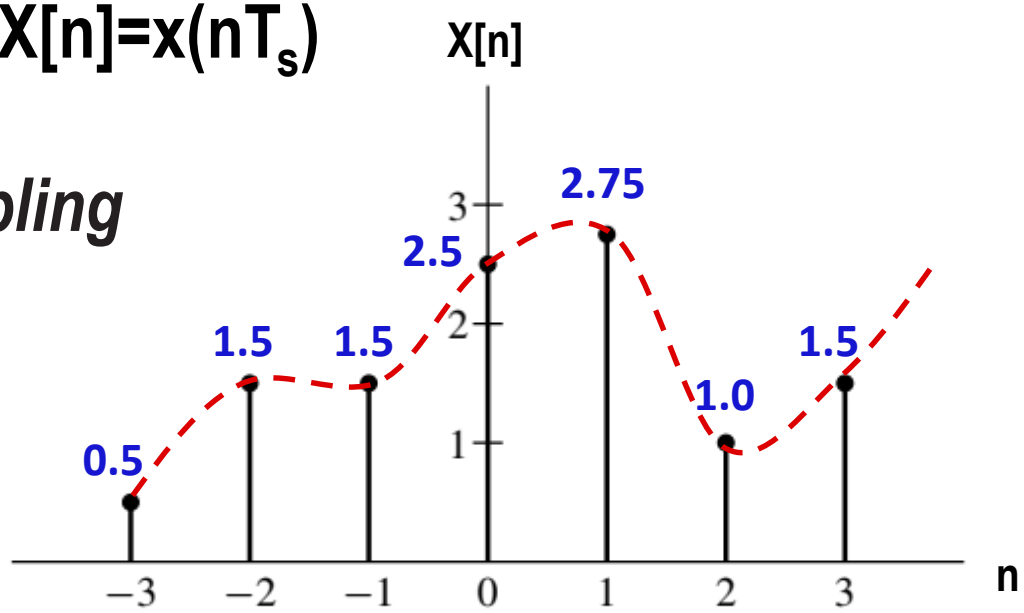
Discrete-Time Signal (1/2)

- Since a discrete-time signal is defined at discrete times, a discrete-time signal is often identified as a *sequence* of numbers, denoted by $x[n]$, here n integer
- Noted: a discrete-time signal $x[n]$ may be obtained by *sampling* a continuous-time signal



$X(t) \Rightarrow X[n]=x(nT_s)$

T_s is the sampling interval

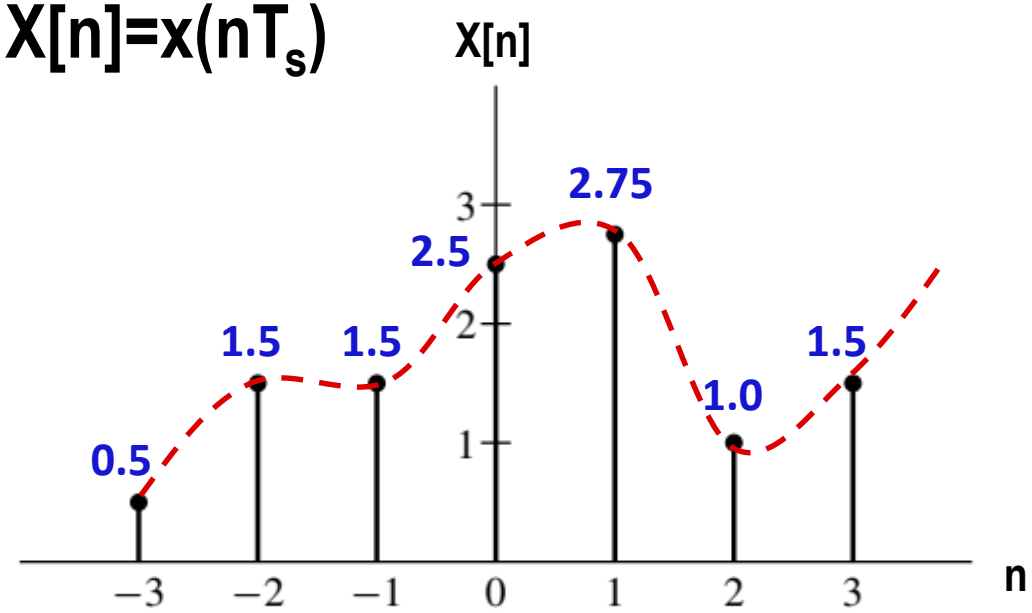


CT ⇒ DT

Discrete-Time Signal (2/2)

- Since a discrete-time signal is defined at discrete times, a discrete-time signal is often identified as a *sequence* of numbers, denoted by $x[n]$, here n integer
- Noted: a discrete-time signal $x[n]$ may be obtained by *sampling* a continuous-time signal

$$X(t) \Rightarrow X[n]=x(nT_s)$$



$$X[n]= [0.5\ 1.5\ 1.5\ \underset{\uparrow}{2.5}\ 2.75\ 1.0\ 1.5\ ]$$

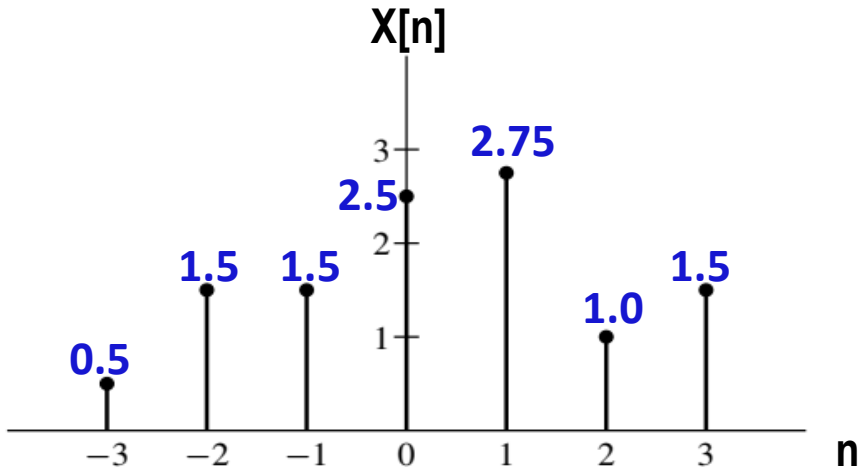
- Conventionally, We use the arrow to denote the $n= 0$ term
- if no arrow is indicated, then the first term corresponds to $n= 0$ and all the values of the sequence are zero for $n < 0$.

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Exercise

- Q1) For a given figure of input signal:-
  - What the is the sequence of x[n],
  - What is the value of x[-3] and x[2]



Ans.

X[n]= [0.5 1.5 1.5 2.5 2.75 1.0 1.5]

n=0

X[-3]= [0.5] and X[2]= [1]

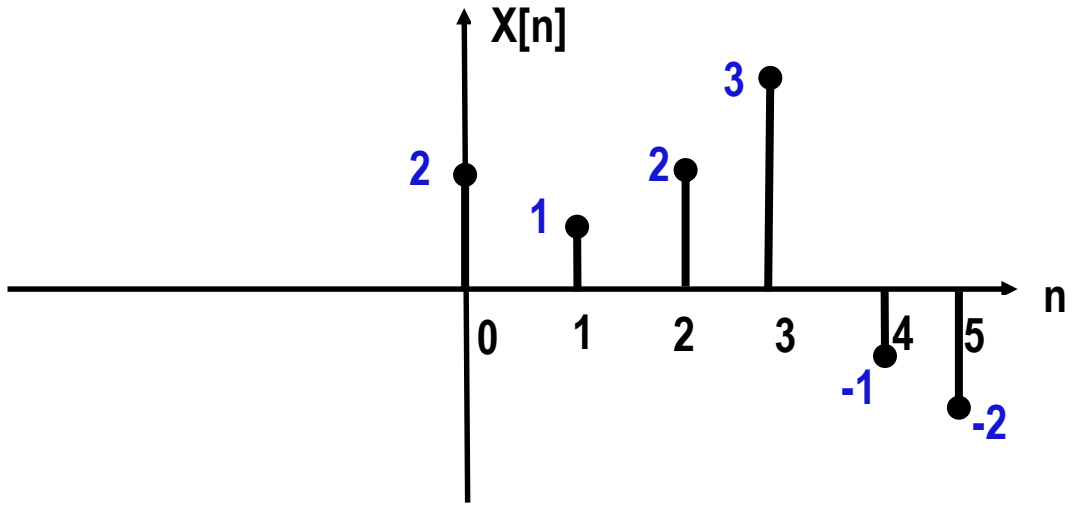
- Q2) For a given sequence of an input signal:-
- X[n]=[2 1 2 3 -1 -2]
  - Plot the input signal in term of x[n] and n

Ans.

X[n]=[2 1 2 3 -1 -2]

n=0

if no arrow is indicated, then the first term corresponds to *n*= 0 and all the values of the sequence are zero for *n* < 0 .



Thanks!